

PREM TIWARI

Software Engineer II · Backend & Data Engineering

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PROFESSIONAL SUMMARY

Software Engineer II with 3.5+ years building real-time IoT and CDC data platforms, including a **\$200K+ annual Databricks cost reduction**, **1,045 ingestion jobs / 2,756 tasks** across US, EU, and AU, and a **45–50 TB/day** OpenSearch pipeline at Smartsheet. Strong across Python, PySpark, Kafka, Flink, Databricks, and AWS.

PROFESSIONAL EXPERIENCE

Software Engineer II — *Smartsheet*

Oct 2025 – Present

- Migrated multi-task streaming jobs from all-purpose to job clusters across US, EU, and AU production, cutting compute costs by **\$200K+ annually**.
- Ran Databricks ingestion across US, EU, and AU: **1,045 jobs / 2,756 tasks** over **11 sources** (MySQL CDC, DynamoDB, Kinesis, Salesforce, NetSuite, Stripe), plus a Snowflake → Delta backfill (**42 jobs / 287 tasks**); deployed with Databricks Asset Bundles.
- CDC from **82 sharded Aurora PostgreSQL and MySQL** databases (Postgres/MySQL → Flink → Kinesis → Delta): **876 jobs / 1,547 tasks**, 282 streaming. A registry expander cut shard onboarding from **~19 configs to 1**.
- Built a high-availability log pipeline (CloudWatch → Kinesis → Logstash → OpenSearch) ingesting **45–50 TB/day** across **50–100 Kinesis streams** and **1,000+ shards**, with **sub-second search**.
- Drove on-call tickets from **~40 to single digits** and cut daily ingestion failures by **~34%**. Built onboarding docs and an access-request dashboard so other teams could self-serve Snowflake, Databricks, AWS, and Jenkins access.

Data Engineer — *Vestas*

Feb 2024 – Sep 2025

- Designed a real-time alarm platform for **~150–200 farms / 19k turbines** on Kafka, Redis, and Kinesis (Docker) — **Protobuf-encoded alarms from Kafka**; built the **frontend API** serving turbine status from DynamoDB with **1–2s lag**, plus a **cursor-paginated file-download API** (DocumentDB + S3); streamed alarms via Kinesis into Delta for analytics.
- Built an OPC-UA IoT pipeline **from scratch** after an AWS Greengrass POC fell short — **22 Kafka partitions** ingesting **~10k data points/sec**, with local-server buffering; enhanced data availability by **96%**.
- Re-built a legacy Databricks streaming job as a Dockerized service (MSK → Kinesis) with **KEDA autoscaling** on Kafka lag, cutting **Delta ingestion latency** from minutes to **10–20 seconds**.
- Optimized Databricks/PySpark pipelines for solar & wind farm data, cutting processing time and latency by **~62%** by switching the source from S3 to Kinesis.

Assistant Data Engineer — *Vestas*

Feb 2023 – Jan 2024

- Built a monitoring system surfacing data latency across source services and pinpointing bottlenecks; reduced API latency by **36%**.
- Automated the previously manual, accuracy-based ML model registration to AWS S3, saving the team **12 hrs/week**; raised unit-test coverage to **95%**.
- Delivered Databricks **LTS** upgrades across **50+** jobs with **data-quality testing** and **zero pipeline failures**.

TECHNICAL SKILLS

Languages: Python, C++, SQL

Data Engineering: Spark, PySpark, Kafka, Flink, Databricks, Airflow, Airbyte, Snowflake, Delta Lake, DBT, ETL, Data Warehousing, Data Modeling, CDC

AWS & Cloud: EC2, S3, Lambda, Kinesis, MSK, SNS, SQS, Glue, API Gateway, VPC, CloudWatch, OpenSearch, Logstash

Databases: Aurora PostgreSQL, Aurora MySQL, PostgreSQL, MongoDB, DynamoDB, Redis

DevOps & Tools: Docker, Kubernetes, KEDA, Terraform, Terragrunt, Ansible, Jenkins, GitHub Actions, Grafana, MLflow

CERTIFICATIONS

- **Snowflake:** Collaboration & Marketplace, Data Application, Data Lake, Data Warehousing, Data Engineering

RECOGNITION

- Public kudos at Smartsheet from Engineering Manager for ownership; recognized by **CTO, Chief AI Officer, SVP**, and senior ICs after leading stretch work equivalent to an **extra quarter** in the current quarter.
- Won an internal Smartsheet hackathon: prototyped **usage analytics** that surface which product capabilities a customer needs from real usage — pitched as a retention and expansion play, reusable across similar accounts.

ACHIEVEMENTS

- **CodeChef**: 5-star, Global Rank 8, College Rank 1 (Starters 70)
- **LeetCode & GeeksforGeeks**: 450+ problems solved
- **TCS CodeVita**: Round 2 Global Rank 83
- **GFG Job-A-Thon 6**: Global Rank 76

EDUCATION

B.Tech, Information Technology — Jabalpur Engineering College, Jabalpur, M.P.

Jul 2019 – Apr 2023